

ROSHAN MOAZED

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EDUCATION

Northeastern University, Boston, MA — M.S. Applied Mathematics (Expected May 2027), GPA: 4.0

University of Massachusetts Boston — B.S. Applied Mathematics, Magna Cum Laude (May 2023)

Brown University — B.A. Creative Writing, Honors (May 2017)

TECHNICAL SKILLS

Languages & Tools: Python, SQL, Excel, VBA, Git, Jupyter, Streamlit

ML / Data Science: XGBoost, scikit-learn, TensorFlow, PyTorch Forecasting, NLP, Recommendation Systems, Computer Vision, RAG, OpenAI API, Pinecone, pandas, NumPy, Hugging Face

Visualization & Analytics: Tableau, Power BI, Plotly

EXPERIENCE

Conservation Resources — Data Analyst (Machine Learning and Predictive Analysis) | Exeter, NH | Aug 2023 – Jan 2026

- Applied K-means clustering, PCA, and time-series analysis to decades of agricultural production, pricing, acreage, and trade data to identify commodity profiles and support portfolio diversification decisions
- Developed supervised machine learning models including linear regression, logistic regression, and random forest for agricultural yield forecasting, achieving ~83% accuracy on apple orchard yield predictions
- Built an internal LLM-powered retrieval chatbot integrating SQL databases with Pinecone vector search to improve access to operational and analytical data
- Delivered predictive modeling and quant. analyses across crop forecasting, asset allocation, and carbon-accounting initiatives using Python and SQL, translating findings into financial and op. insights for leadership

AI Lean — Customer Satisfaction Analyst | Boston, MA | Dec 2022 – Jun 2023

- Built dashboards and reports on customer service performance and satisfaction across multiple datasets.
 - Reduced average ticket response time by 47% and resolution time by 54% through workflow analytics.
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PROJECTS

Computer Vision for Brain Hemorrhage Segmentation & Classification

- Developed TensorFlow computer-vision pipeline for CT hemorrhage segmentation using U-Net/ResU-Net, ROI cropping, active learning, and multi-task classification; improved Dice coefficient from ~0.67 to 0.76
- Built and evaluated segmentation/classification models using Dice, IoU, BCE/Tversky losses, data augmentation, and engineered spatial features; diagnosed data leakage and class collapse across model iterations

Memory-Augmented Conversational AI System

- Designed and evaluated a memory-augmented chatbot using semantic similarity, SVM, LLM-based extraction, Pinecone retrieval, and conflict-aware memory updates; improved retrieval accuracy from 60% to 80%

Retail Demand Forecasting & Inventory Optimization

- Built XGBoost demand-forecasting and inventory-planning system with engineered temporal/business features and Streamlit deployment

Relevant Coursework: Machine Learning, Mathematical Modeling, Applied Linear Algebra, Natural Language Processing | Upcoming: Machine Learning II, Probability, Applied Statistics

Interests: Machine Learning, Computer Vision, Optimization, Statistical Modeling, NLP & LLMs